Development Plan Document

University Internship & Industry Matching Portal

1. Introduction

1.1 Purpose

This document provides a detailed and comprehensive development plan for the University Internship & Industry Matching Portal. It is intended for the development team, project supervisors, and stakeholders to outline tasks, responsibilities, timelines, and workflow for the successful completion of the project.

1.2 Scope

The plan covers all stages of development including design, frontend and backend implementation, database creation, project and competition modules, testing, deployment, and documentation.

2. Team Structure and Responsibilities

The project has a team of 4 members:

Member	Role	Responsibilities
Member 1	Backend Lead	Design database, develop ASP.NET Core APIs, implement business logic, authentication, and internship/project workflows
Member 2	Frontend Lead	React.js development, Student Portal UI, Dashboard, Form validations, API integration, Responsive design
Member 3	Company & Admin Modules Developer	Implement Company portal, CV review, Interview scheduling, Admin dashboard, Role management, Approval workflows
Member 4	DevOps & Project/Competition	Azure deployment, CI/CD pipeline, Document storage integration, Project &

Member	Role	Responsibilities
	Modules Developer	Competition modules, Monitoring and deployment support

3. Development Methodology

- **Agile methodology**
- **Sprint-based development** with weekly reviews
- Continuous integration and testing
- Iterative delivery with feedback loops

4. Technology Stack

- **Frontend:** React.js
- **Backend:** ASP.NET Core Web API
- **Database:** MySQL
- **Cloud Deployment:** Microsoft Azure
- **Storage:** Azure Blob Storage
- **Authentication & Security:** JWT, Role-based access control

5. Development Phases and Timeline

Total Duration: 16 Weeks

Phase 1 – Planning & Design (Weeks 1–2)

- Finalize requirements and scope
- Prepare SRS, Project Proposal, and System Architecture
- Design database schema and ER diagrams
- Plan UI wireframes

Deliverables: SRS Document, Architecture Design, Wireframes, Database Schema

Phase 2 – Backend Core Development (Weeks 3–6)

- Set up ASP.NET Core project structure
- Develop authentication and authorization modules
- Internship module APIs
- Project and competition module APIs
- Application tracking and status management APIs
- Resume and document management APIs

Deliverables: Functional backend APIs, Database connected, Basic testing

Phase 3 – Frontend Development (Weeks 5–9)

- Student Portal implementation
- Internship browsing and application pages
- Project and competition pages
- Dashboard for students
- Integration with backend APIs

Deliverables: Student Portal functional prototype

Phase 4 – Company & Admin Modules (Weeks 8–11)

- Company portal UI & functionality
- CV review, accept/reject, and interview scheduling
- Admin dashboard, role management, and approvals
- Integration with backend

Deliverables: Company & Admin Portals functional, API fully integrated

Phase 5 – Project & Competition Modules (Weeks 10–13)

- Project creation and participation modules
- Competition discovery and registration modules
- Leader and student participation workflows

- Notifications and status updates

Deliverables: Project & Competition modules fully functional

Phase 6 – Testing & Quality Assurance (Weeks 12–14)

- Unit testing (frontend & backend)
- Integration testing
- System testing
- Security testing
- Performance testing

Deliverables: Fully tested system, Bug reports, QA approval

Phase 7 – Deployment & Optimization (Weeks 14–15)

- Azure deployment for frontend and backend
- Database hosting on Azure MySQL
- Document storage setup (Azure Blob Storage)
- Configure CI/CD pipeline (Azure DevOps/GitHub Actions)
- Production testing and optimization

Deliverables: Live system on Azure, fully functional, accessible

Phase 8 – Documentation & Final Review (Week 16)

- Technical documentation
- User manuals for students, companies, and admin
- Deployment guides
- Final demonstration to stakeholders

Deliverables: Complete project documentation, final system handover

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6. Risk Management

Risk	Mitigation
Integration delays	Early backend API completion, parallel development sprints
Deployment configuration issues	Test Azure deployment in staging environment, use CI/CD pipelines
Time overruns	Track progress weekly, adjust sprints, prioritize core modules
Security vulnerabilities	Input validation, role-based access control, HTTPS, encryption

7. Quality Assurance Strategy

- Continuous integration testing
- Code reviews for each module
- Automated and manual test cases
- Bug tracking and resolution
- Performance monitoring on Azure

8. Development Tools

- Version control: Git/GitHub
- Frontend: Node.js, npm, React.js
- Backend: Visual Studio, ASP.NET Core
- Database: MySQL Workbench
- Cloud: Microsoft Azure (App Service, Database, Blob Storage)
- Testing: Postman, Selenium, Jest (frontend)
- CI/CD: Azure DevOps / GitHub Actions

9. Deliverables

- Functional web application (Student, Company, Admin portals)

- Azure deployed system
- Backend APIs
- Database schema and ER diagram
- UI wireframes
- Technical and user documentation
- Test cases and QA reports
- Deployment guides

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10. Monitoring & Maintenance Plan

- Monitor system performance using Azure Application Insights
- Regular backups of database and documents
- Update system with patches and security updates
- Monitor user feedback and resolve issues